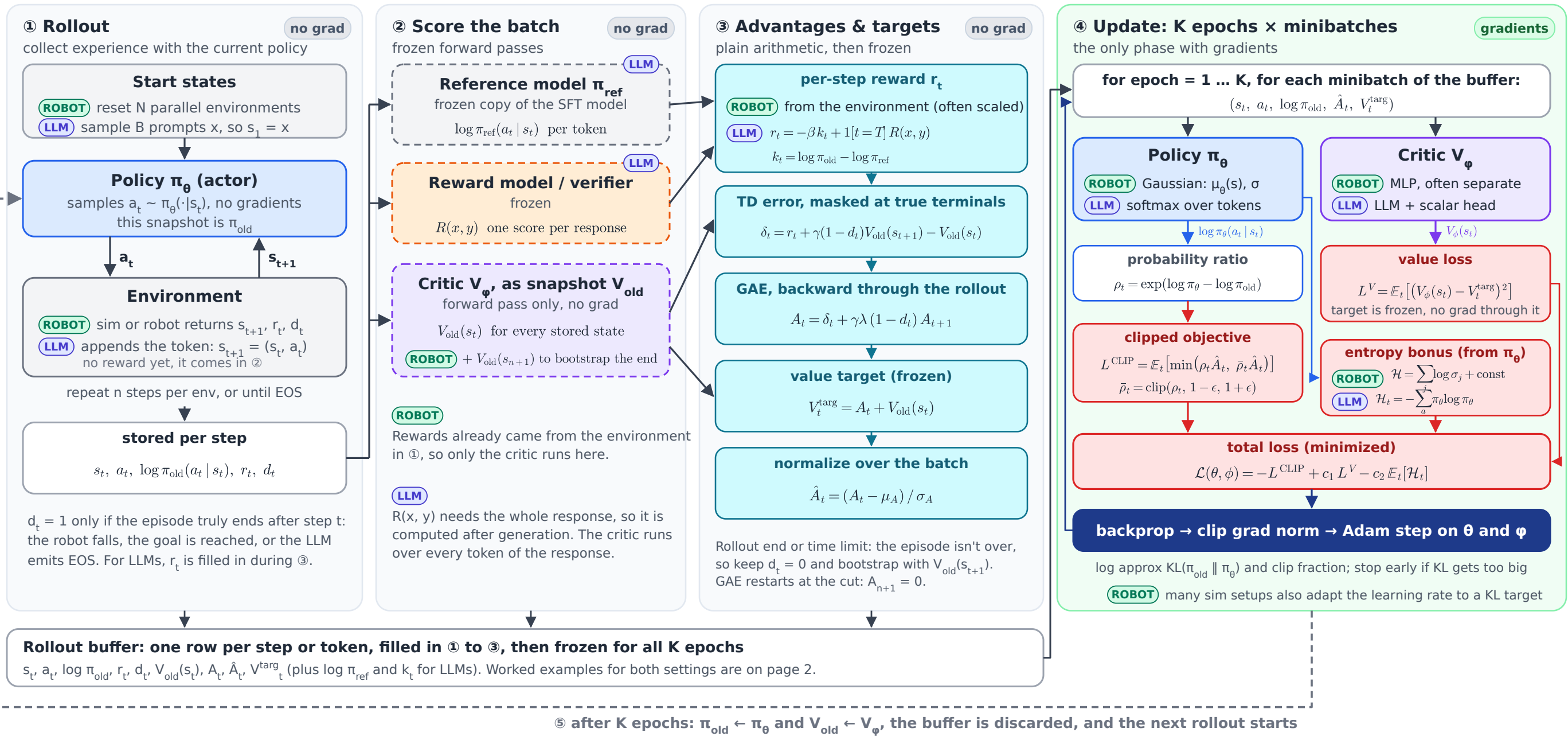


PPO cheat sheet · generic, with robotics and LLM variants

One training iteration: what each part does, what gets computed, what stays frozen, and where gradients flow. Untagged parts are identical in both settings.

trainable model
frozen model or snapshot
arithmetic, no grad
loss term
no-gradient phase
gradient phase
ROBOT robotics / control only
LLM LLM only



Pseudocode

```

for iteration = 1, 2, ...:
    rollout with  $\pi_\theta$  (this snapshot is  $\pi_{old}$ ), no grad:
        ROBOT: N envs x n steps →  $s_t, a_t, r_t, d_t, \log \pi_{old}$ 
        LLM: B prompts → full responses →  $s_t, a_t, \log \pi_{old}$ 
         $V_{old}(s_t) \leftarrow$  critic snapshot (ROBOT: also  $V_{old}(s_{n+1})$ )
        LLM:  $R(x, y), \log \pi_{ref} \rightarrow r_t = -\beta \cdot k_t + [t=T] \cdot R$ 
         $\delta_t, A_t$  (GAE, masked by  $d_t$ ),  $V^{targ}_t, \hat{A}_t$  # frozen
    for epoch in 1..K:
        for minibatch in shuffle(buffer):
             $\rho_t = \exp(\log \pi_\theta - \log \pi_{old})$ 
             $\text{loss} = -L^{CLIP} + c_1 \cdot L^V - c_2 \cdot H$ 
            backprop; clip grad norm; Adam step ( $\theta, \phi$ )
        if approx_KL( $\pi_{old} \parallel \pi_\theta$ ) > target: break
    discard buffer # next:  $\pi_{old} \leftarrow \pi_\theta, V_{old} \leftarrow V_\phi$ 
    
```

Key facts

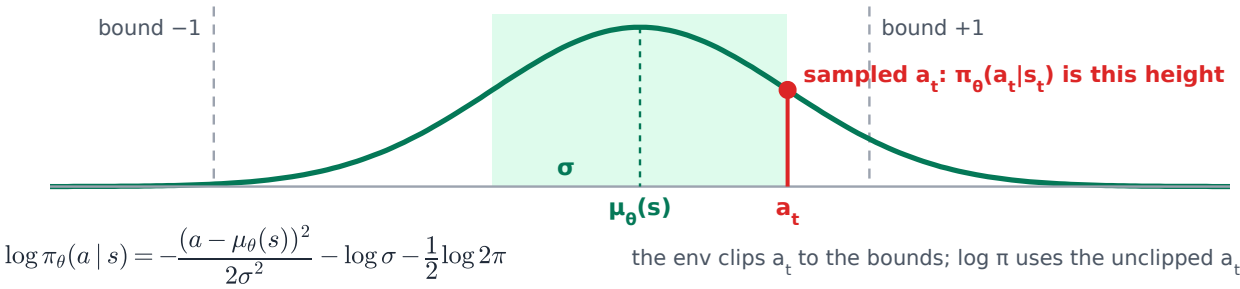
- Frozen for the whole batch: $\log \pi_{old}, \hat{A}_t$ and V^{targ}_t . Only θ and ϕ change inside ④.
- First minibatch step: $\rho_t = 1$, so the gradient is exactly REINFORCE with advantage $\hat{A}_t$.
- Clipping zeroes a sample's gradient only on the helpful side ($\hat{A} > 0, \rho > 1 + \epsilon$ or $\hat{A} < 0, \rho < 1 - \epsilon$).
- d_t cuts GAE at true terminals. Rollout ends and time limits aren't terminals: bootstrap.
- ROBOT: dense rewards, $\gamma = 0.99$, cheap steps, ~10 epochs per batch; σ drives exploration.
- LLM: one sparse score + KL to π_{ref} , $\gamma = 1$, costly generation, 1 to 4 epochs per batch.
- LLM: GRPO keeps L^{CLIP} but drops the critic: advantage = group-normalized final reward.

Robotics vs LLM, side by side

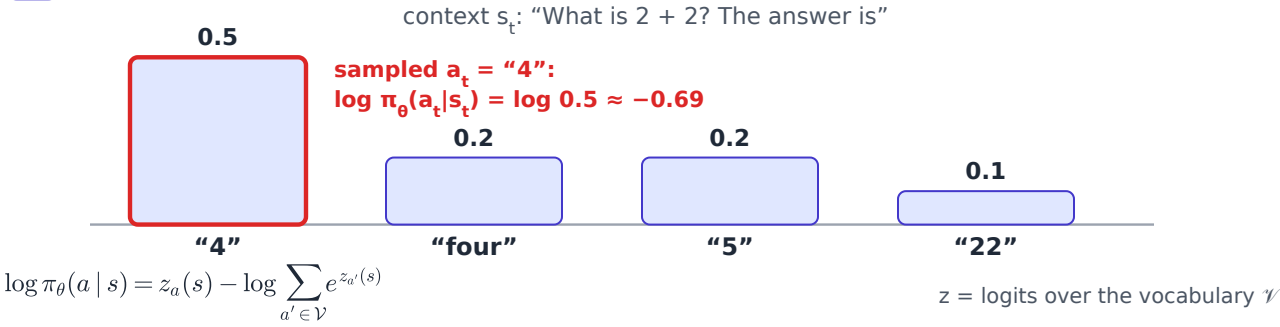
Same algorithm, different plumbing. These are the places where implementations actually differ.

	ROBOT Robotics / continuous control	LLM LLM fine-tuning (RLHF, RLVR)
State s_t	Observation vector: joint angles, velocities, contacts (sometimes images)	Prompt plus tokens so far: $s_t = (x, y_{<t})$
Action a_t	Continuous vector in $\mathbb{R}^d$: joint position targets or torques	One token from a vocabulary of roughly 10^5
Policy head	Gaussian: mean $\mu_\theta(s)$ from an MLP, learned $\log \sigma$ (often state-independent)	Softmax over the vocabulary logits
$\log \pi_\theta(a_t \mid s_t)$	Gaussian log-density, summed over the d action dimensions	Log-softmax of the sampled token
Environment step	Simulator or real robot returns s_{t+1}, r_t, d_t ; stochastic, costly on hardware	Deterministic: append the token to the context
Reward	Dense and shaped every step: progress, energy, posture, contact penalties	Sparse: one score $R(x, y)$ at the end, plus a per-token KL penalty
Episode end	Termination (fall, goal) or a time limit, e.g. 1000 steps	End-of-sequence token or the maximum length
Rollout	N parallel envs $\times$ n steps, cutting across episode boundaries	B prompts with complete responses, often several per prompt
Bootstrapping	At the rollout's last step and at time-limit truncation	Not needed: every response is a complete episode
Discount γ	0.99, an effective horizon of roughly 100 steps	1, no discounting within a response
Anchor	Usually none	KL to π_{ref} : in the reward (PPO) or in the loss (GRPO)
Critic	Small MLP, usually a separate network from the policy	LLM-sized, often initialized from the reward model; GRPO drops it
Normalization	Running observation normalization; rewards scaled by a running return std	Advantage whitening; reward normalization is common
Exploration	Gaussian noise set by σ ; entropy bonus c_2 from 0 to 0.01	Sampling temperature; entropy bonus usually 0
Epochs per batch	Many: 10 in the PPO paper's MuJoCo runs	Few: 1 to 4, because samples are expensive
Bottleneck	Environment steps, in simulation or on hardware	Generating responses (inference)

ROBOT Gaussian policy head (one action dimension shown)



LLM Softmax policy head (top tokens shown)



ROBOT Worked example: 4 steps from one env ($\gamma = 0.99$, $\lambda = 0.95$)

	r_t	d_t	$V_{\text{old}}(s_t)$	δ_t	A_t	V_t^{targ}
$t = 1$	0.50	0	2.00	-0.71	-2.40	-0.40
$t = 2$	-1.00	1	0.80	-1.80	-1.80	-1.00
$t = 3$	0.40	0	1.50	0.58	1.22	2.72
$t = 4$	0.60	0	1.70	0.68	0.68	2.38

$t = 2$: the robot falls ($d_2 = 1$), so the GAE chain is cut and $V_2^{\text{targ}} = r_2$.
 $t = 3$ starts a new episode after the reset.
 $t = 4$: the rollout ends mid-episode, so δ_4 bootstraps with $V_{\text{old}}(s_5) = 1.80$:
 $\delta_4 = 0.60 + 0.99 \times 1.80 - 1.70 = 0.68$, and $A_4 = \delta_4$ because $A_5 = 0$ at the cut.

LLM Worked example: 3-token correct answer ($\beta = 0.1$, $\gamma = 1$, $\lambda = 0.95$)

	a_t	$\log \pi_{\text{old}}$	$\log \pi_{\text{ref}}$	k_t	r_t	V_{old}	A_t	V_t^{targ}
$t = 1$	"It"	-1.20	-1.30	0.10	-0.01	0.50	0.43	0.93
$t = 2$	"is"	-0.50	-0.70	0.20	-0.02	0.60	0.36	0.96
$t = T = 3$	"4"	-0.10	-0.20	0.10	0.99	0.80	0.19	0.99

$R(x, y) = 1$ enters only at $t = T$: $r_3 = -0.01 + 1 = 0.99$. Earlier r_t are just $-\beta k_t$.
 $d_T = 1$ at EOS, so $V_{\text{old}}(s_4) = 0$ and $V_3^{\text{targ}} = r_3$. Nothing needs bootstrapping.
With $\gamma = 1$ and $\lambda = 0.95$, early-token targets are mostly the real final reward here;
on responses with thousands of tokens they become mostly bootstrapped from V_{old} .

ROBOT **LLM** Hybrid: VLA policies such as OpenVLA discretize each action dimension into 256 bins and emit them as tokens. Their $\log \pi$ follows the LLM column; rewards and episodes follow the robotics column.

Formulas

Generic unless tagged. ρ_t is the probability ratio (papers often write $r_t(\theta)$, which clashes with the reward r_t). $\mathbb{E}_t$ averages over all steps or tokens in the batch or minibatch.

1 • Setup

state, action, reward	$s_t, \quad a_t, \quad r_t, \quad t = 1, \dots, n$ (or T)
done flag	$d_t = 1$ if the episode truly terminates after step t , else 0
ROBOT rollout	N envs $\times$ n steps, cutting across episodes
LLM rollout	B prompts; $y = (a_1, \dots, a_T), \quad s_t = (x, a_{<t})$
policies	π_θ (trained), π_{old} (sampled the batch), π_{ref} (LLM: frozen SFT)
critic	V_ϕ (trained), V_{old} (its snapshot on the batch)
hyperparameters	$\epsilon, \gamma, \lambda, c_1, c_2, K; \quad \beta$ (LLM KL coefficient)

2 • Per-step reward

ROBOT reward	$r_t = r(s_t, a_t)$ (often divided by a running std of the return)
LLM KL estimate	$k_t = \log \pi_{\text{old}}(a_t \mid s_t) - \log \pi_{\text{ref}}(a_t \mid s_t)$
LLM expectation	$\mathbb{E}_{a_t \sim \pi_{\text{old}}}[k_t] = D_{\text{KL}}(\pi_{\text{old}}(\cdot \mid s_t) \parallel \pi_{\text{ref}}(\cdot \mid s_t))$
LLM reward	$r_t = -\beta k_t + 1[t = T] R(x, y)$
ROBOT rewards are dense and shaped. LLM rewards are sparse: the score only enters at the last token.	

3 • Advantages and value targets (GAE)

TD error	$\delta_t = r_t + \gamma(1 - d_t) V_{\text{old}}(s_{t+1}) - V_{\text{old}}(s_t)$
GAE, backward	$A_t = \delta_t + \gamma \lambda (1 - d_t) A_{t+1}, \quad A_{n+1} = 0$ at the cut
GAE, summed	$A_t = \sum_{l \geq 0} (\gamma \lambda)^l \delta_{t+l}$ (until the episode or rollout ends)
value target	$V_t^{\text{targ}} = A_t + V_{\text{old}}(s_t)$
= λ -return	$V_t^{\text{targ}} = (1 - \lambda) \sum_{n \geq 1} \lambda^{n-1} G_t^{(n)}$
n-step return	$G_t^{(n)} = \sum_{l=0}^{n-1} \gamma^l r_{t+l} + \gamma^n V_{\text{old}}(s_{t+n})$
limits	$\lambda = 1$: $A_t = G_t - V_{\text{old}}(s_t)$ (MC), $\lambda = 0$: $A_t = \delta_t$ (TD)
bootstrap rule	rollout end or time limit: $d_t = 0$, use $V_{\text{old}}(s_{t+1})$; true terminal: $d_t = 1$
normalize	$\hat{A}_t = (A_t - \mu_A) / (\sigma_A + 10^{-8})$ over the batch
Past an episode's end, $G^{(n)}$ stops growing: it equals the full return (bootstrapped with V_{old} at a cut). LLM: each response is a complete episode ending in EOS, so $d_T = 1$, $\gamma = 1$, and nothing is bootstrapped.	

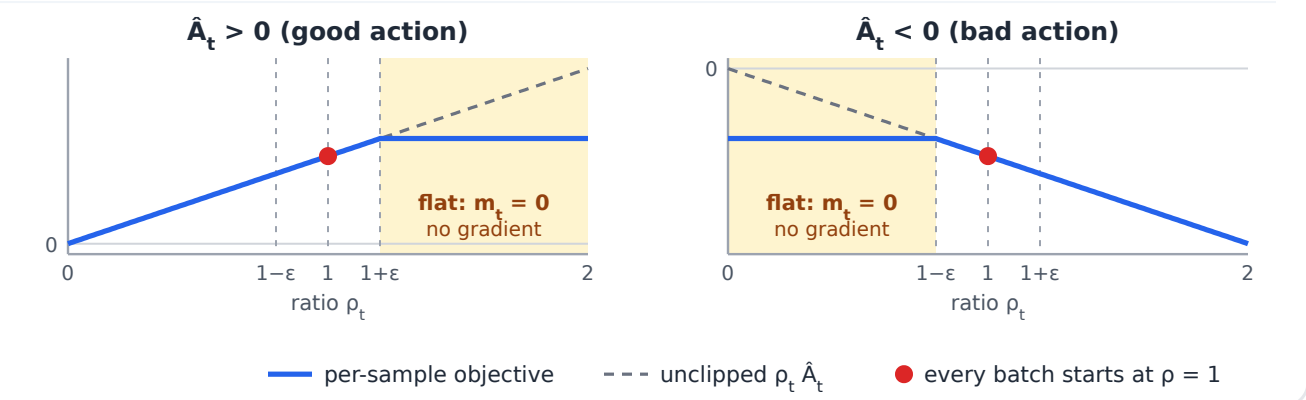
4 • Policy objective (clipped surrogate)

probability ratio	$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\text{old}}(a_t \mid s_t)} = \exp(\log \pi_\theta - \log \pi_{\text{old}})$
unclipped surrogate	$L^{\text{CPI}}(\theta) = \mathbb{E}_t[\rho_t \hat{A}_t]$
clipped surrogate	$L^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$
its gradient	$\nabla_\theta L^{\text{CLIP}} = \mathbb{E}_t[m_t \hat{A}_t \rho_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)]$
mask	$m_t = 0$ if $\hat{A}_t > 0, \rho_t > 1 + \epsilon$ or $\hat{A}_t < 0, \rho_t < 1 - \epsilon$; else $m_t = 1$
first step ($\rho = 1$)	$\nabla_\theta L^{\text{CLIP}} = \mathbb{E}_t[\hat{A}_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)]$ (REINFORCE)
log-derivative trick	$\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$

5 • Critic (value) loss

value loss	$L^V(\phi) = \mathbb{E}_t[(V_\phi(s_t) - V_t^{\text{targ}})^2]$ (often $\times \frac{1}{2}$)
clipped variant (optional)	$L_{\text{clip}}^V = \mathbb{E}_t[\max((V_\phi - V_t^{\text{targ}})^2, (V_{\text{old}} + \text{clip}(V_\phi - V_{\text{old}}, -\epsilon_v, \epsilon_v) - V_t^{\text{targ}})^2)]$
V_t^{targ} is a frozen number: gradients flow only through $V_\phi(s_t)$.	

6 • The clip, per sample ($\epsilon = 0.2$)



7 • Policy distributions

ROBOT Gaussian head	$a \sim \mathcal{N}(\mu_\theta(s), \text{diag}(\sigma_1^2, \dots, \sigma_d^2)), \quad a \in \mathbb{R}^d$
ROBOT log-probability, summed over the d action dimensions	$\log \pi_\theta(a \mid s) = -\sum_{j=1}^d \left[\frac{(a_j - \mu_{\theta,j}(s))^2}{2\sigma_j^2} + \log \sigma_j \right] - \frac{d}{2} \log 2\pi$
ROBOT entropy	$\mathcal{H} = \sum_{j=1}^d \log \sigma_j + \frac{d}{2} (1 + \log 2\pi)$
LLM softmax head	$\log \pi_\theta(a \mid s) = z_a(s) - \log \sum_{a' \in \mathcal{V}} e^{z_{a'}(s)}$
LLM entropy	$\mathcal{H}_t = -\sum_{a \in \mathcal{V}} \pi_\theta(a \mid s_t) \log \pi_\theta(a \mid s_t)$
ROBOT: the env clips actions to their bounds, but log π uses the unclipped sample. The learned σ sets how much the policy explores. LLM: z are the logits over the vocabulary $\mathcal{V}$; a response's log-probability sums over its tokens.	

8 • Total loss

total (minimized)	$\mathcal{L}(\theta, \phi) = -L^{\text{CLIP}}(\theta) + c_1 L^V(\phi) - c_2 \mathbb{E}_t[\mathcal{H}_t]$
The PPO paper writes the same thing as maximizing $L^{\text{CLIP}} - c_1 L^V + c_2 \mathcal{H}$. With separate policy and critic networks, c_1 only rescales the critic's step size.	

9 • Diagnostics, logged every epoch

approx KL (k1)	$\widehat{D}_{\text{KL}}(\pi_{\text{old}} \parallel \pi_\theta) = \mathbb{E}_t[-\log \rho_t]$
approx KL (k3)	$\mathbb{E}_t[(\rho_t - 1) - \log \rho_t]$ (unbiased, every term ≥ 0)
clip fraction	$\mathbb{E}_t[1(\rho_t - 1 > \epsilon)]$
explained variance	$1 - \text{Var}(V^{\text{targ}} - V_{\text{old}}) / \text{Var}(V^{\text{targ}})$
Early stopping: end the epoch loop if approx KL exceeds a target (e.g. 0.01).	

10 • Common starting values (tune for your setup)

both	$\epsilon = 0.2, \quad \lambda = 0.95$
ROBOT PPO paper (MuJoCo)	$\gamma = 0.99, K = 10$, minibatch 64, horizon 2048, Adam 3×10^{-4}
ROBOT loss weights	$c_1 \approx 0.5$ to 1, c_2 from 0 to 0.01
LLM typical	$\gamma = 1, K = 1$ to 4, $c_2 = 0$, β fixed or adapted to a KL target
ROBOT: massively parallel simulators often run thousands of envs with short rollouts per env and adapt the learning rate to keep approx KL near a target.	

LLM 11 • GRPO, for comparison

group advantage	$\hat{A}_{i,t} = \frac{R_i - \text{mean}(R_1, \dots, R_G)}{\text{std}(R_1, \dots, R_G)}$ for every token of y_i
objective (G responses per prompt)	$J(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{ y_i } \sum_{t=1}^{ y_i } (\min(\rho_{i,t} \hat{A}_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t}) - \beta D_{i,t}) \right]$
KL term (k3)	$D_{i,t} = \pi_{\text{ref}} / \pi_\theta - \log(\pi_{\text{ref}} / \pi_\theta) - 1 \geq 0$
No critic and no GAE: the group mean plays the role of V . Dr. GRPO drops the std and the $1/ y_i $ terms. With one update per batch, $\rho = 1$ and this is REINFORCE with a group baseline.	